SOURADIP NATH

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Research Interests

AI Security; Human Factors in Security and Privacy; Human-AI Collaboration; AI Governance and Access Control

Education

- **Arizona State University** — *Fall 2023 - Spring 2028 (Expected)*
  Doctor of Philosophy (Ph.D.) in Computer Science — Tempe, AZ, USA
  - Advisor: Gail-Joon Ahn
- **Indian Institute of Engineering Science and Technology, Shibpur** — *Spring 2022*
  Bachelor of Technology (B.Tech.) in Information Technology — Howrah, India
  - GPA: 9.93/10.0 (First Class with Honors, Presidential Gold Medalist)

Publications

[12] Ananta Soneji, **Souradip Nath**, Moritz Schloegel, Yan Shoshitaishvili, Gail-Joon Ahn, Adam Doupe, and Carlos Rubio-Medrano. "Beyond the Buzzword: How do Professionals Understand and Translate Zero Trust?" To appear in *ACM SIGSAC Conference on Computer and Communications Security (CCS)*, November 2026.

[11] **Souradip Nath**, Chih-Yi Huang, Aditi Ganapathi, Kashyap Thimmaraju, Jaron Mink, Gail-Joon Ahn "Like a Hammer, It Can Build, It Can Break: Large Language Model Uses, Perceptions, and Adoption in Cybersecurity Operations on Reddit", To appear in *Symposium on Usable Privacy and Security (SOUPS)*, August 2026.

[10] Carlos Rubio-Medrano, **Souradip Nath**, Ananta Soneji, Jennifer Mondragon, Jaejong Baek, and Gail-Joon Ahn. "Towards Agentic AI for Access Control in Cyber-infrastructures: Exploring the Security and Human Factors", To appear in *31st ACM Symposium on Access Control Models and Technologies (SACMAT)*, July 2026.

[9] Sukwha Kyung, **Souradip Nath**, Jaejong Baek, and Gail-Joon Ahn. "Deception by Design: A Configurable Platform for Flexible Cyber Deception Strategy Testing and Evaluation", To appear in *24th International Conference on Applied Cryptography and Network Security (ACNS), June 2026*. [PDF]

[8] Hui Jun Tay, **Souradip Nath**, Arvind S Raj, Abhay Bhat, Ishan Bansal, Audrey Dutcher, Moritz Schloegel, Adam Doupé, Tiffany Bao, Yan Shoshitaishvili, and Ruyou Wang. "Responsible Disclosure is a Two-Way Street: Empirically Measuring the Responsible Disclosure Contract in the Firmware Ecosystem", To appear in *2026 IEEE Symposium on Security and Privacy (SP)*, May 2026. [PDF]
Distinguished Paper Award

[7] Kashyap Thimmaraju, Duc Anh Hoang, **Souradip Nath**, Jaron Mink, Gail-Joon Ahn. "Before the Vicious Cycle Starts: Preventing Burnout Across SOC Roles Through Flow-Aligned Design", in the *Workshop on SOC Operations and Construction (WOSOC)*, 2026. [PDF]

[6] Zehua Zhang, Ati Priya Bajaj, Divij Handa, Siyu Liu, Arvind S Raj, Hongkai Chen, Hulin Wang, Yibo Liu, Zion Leonahenahe Basque, **Souradip Nath**, Vishal Juneja, Nikhil Chapre, Yan Shoshitaishvili, Adam Doupé, Chitta Baral, and Ruoyu Wang, "BuildBench: Benchmarking LLM Agents on Compiling Real-World Open-Source Software," in NeurIPS 4th Deep Learning for Code Workshop, 2025. [PDF]

[5] **Souradip Nath**, Ananta Soneji, Jaejong Baek, Carlos Rubio-Medrano and Gail-Joon Ahn. "Towards Collaboration-Aware Resource Sharing in Research Computing Infrastructures", in *2025 IEEE International Conference on Collaboration and Internet Computing (CIC)*. [PDF]

[4] Sukwha Kyung, **Souradip Nath**, Jaejong Baek, and Gail-Joon Ahn. "Harnessing Software-Defined Network for Cyber Deception Modeling and Analysis", in *2025 IEEE Conference on Network Function Virtualization and Software Defined Networks (NFV-SDN)*. [PDF]

[3] **Souradip Nath**, Ananta Soneji, Jaejong Baek, Tiffany Bao, Adam Doupé, Carlos Rubio-Medrano, and Gail-Joon Ahn, "It's almost like Frankenstein: Investigating the Complexities of Scientific Collaboration and Privilege Management within Research Computing Infrastructures", in *2025 IEEE Symposium on Security and Privacy (SP)*, May 2025, pp. 2995–3013. [PDF]

[2] **Souradip Nath**, Keb Summers, Jaejong Baek, and Gail-Joon Ahn. "Digital Evidence Chain of Custody: Navigating New Realities of Digital Forensics." In *2024 IEEE 6th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)*, pp. 11-20. [PDF]

[1] **Souradip Nath**, and Ruchira Naskar. "Automated image splicing detection using deep CNN-learned features and ANN-based classifier." *Signal, Image and Video Processing* 15, no. 7 (2021): 1601-1608. [PDF]

EXPERIENCE

- **Global Security Initiative, Arizona State University: Graduate Research Associate** *Spring 2023 - Current*
  Advisor: Gail-Joon Ahn Tempe, AZ, USA
  - Leading research projects that integrate human-centered and technical approaches to security and privacy.
  - Designing and executing semi-structured interview studies and conducting thematic analysis to understand user practices, challenges, and needs around security and privacy.
  - Complementing human-centered findings with technical system-level exploration to design and implement deployable, system-level access control solutions.
  - Synthesizing research findings into actionable insights and disseminating them to diverse audiences through papers, posters and conference talks.
- **Deutsche Bank Technology, Data & Innovation: Technical Analyst Intern** *Summer 2021*
  Manager: Pragnya Seth Pune, India
  - Utilized Python Pandas library to clean large datasets of scheduled jobs, conducted data analysis, and came up with an optimization algorithm for job scheduling using the Greedy algorithm.
  - Collaborated with a team of 3 and proposed a general framework for identifying an optimized job scheduling against a multi-parameter cost function.

TEACHING EXPERIENCE

- **School of Computing and Augmented Intelligence, ASU: Teaching Assistant** *Fall 2024*
  Course: Computer and Network Forensics, Instructor: Jaejong Baek Tempe, AZ, USA
  - Assisted in delivering a senior-level course (lectures and lab sessions) on Digital Forensics, covering the fundamentals of computer and network forensics including cloud, email, and mobile forensics.

PROJECTS

- **(Published) Exploring LLM Uses, Perceptions, and Adoption in Cybersecurity Operations on Reddit**
  Tools: Python PRAW, Reddit API (data collection), MaxQDA, OpenAI API (qualitative analysis), R (quantitative analysis)
  - Conducted a **mixed-methods Reddit-based study** combining qualitative thematic analysis and quantitative tests to explore uses, perception, and adoption of security practitioners regarding LLM-augmented SOCs.
  - Surfaced several **previously unexplored aspects**, including the fragmented ecosystem of security-specific LLM tools, varying levels of autonomy in LLM-assisted workflows, practitioner curiosity and skepticism, and workforce implications in increasingly AI-augmented SOCs.
  - Identified **research opportunities** surrounding trustworthy AI-assisted workflows, human-AI co-learning and the long-term sustainability of the SOC workforce.
- **(Published) A Human-centered Exploration of Access Control within Scientific Collaboration**
  Tools: Zoom (data collection), Otter.ai (transcription), MaxQDA (qualitative analysis), ReCal2 (intercoder reliability)
  - Conducted a **semi-structure interview-based study** with 24 key stakeholders of Research Computing Infrastructures (RCIs) to identify security and usability challenges in access control for scientific collaboration.
  - Applied qualitative **thematic analysis** to interview data, uncovering that informal, trust-based access control and fragmented system design create significant security challenges and usability issues.
  - Formulated actionable recommendations for improving security and usability in RCIs including design heuristics for effective access control, establishing a new direction for future research in RCI security.
- **(Published) Towards Collaboration-Aware Resource Sharing in Research Computing Infrastructures**
  Tools: Python (tool implementation), VirtualBox, Ubuntu Server, OpenLDAP, NFS, Slurm (environment simulation)
  - Investigated existing resource sharing practices within Research Computing Infrastructures (RCIs), revealing key gaps in their ability to support secure authorization in dynamic collaborative workflows.
  - Developed CLEARS, a novel framework for collaboration-aware resource sharing that formally represents collaboration contexts to guide secure, dynamic access authorization in RCIs.
  - Implemented a prototype of CLEARS and conducted a scenario-based case study with experimental evaluation to demonstrate that it delivers precise access enforcement while maintaining minimal execution overhead.

PROFESSIONAL SERVICE

- **Organizing Committee Member:** ACM SACMAT 2025-2026
- **Technical Program Committee Member:** MetaCRiSP 2026 (IEEE S&P Workshop)
- **Artifact Evaluation Committee Member:** ACM SACMAT 2026
- **External Reviewer:** IEEE ICDM 2026, ACM ASIACCS 2025, ACM CODASPY 2025, ACM SACMAT 2024-2025, CSET 2024

Honors and Awards

- **Graduate Student Government (GSG) Travel Grant – USD 950** *July 2026*
  Arizona State University, AZ, USA
- **Student Travel Grant – USD 700** *November 2025*
  11th IEEE International Conference on Collaboration and Internet Computing, Pittsburgh, PA, USA
- **President of India Gold Medal for University Topper (Undergraduate)** *December 2022*
  Indian Institute of Engineering Science and Technology, Shibpur, India
- **Silver Medal for Department Topper (Information Technology)** *December 2022*
  Indian Institute of Engineering Science and Technology, Shibpur, India
- **New American University Scholarship – USD 10,000** *August 2022*
  Arizona State University, AZ, USA []
- **Swami Vivekananda Merit-cum-Means Scholarship – INR 2,40,000** *August 2018*
  Government of West Bengal, India []
- **GP Birla Merit-cum-Means Scholarship – INR 2,00,000** *August 2018*
  GP Birla Educational Foundation, India []

Mentorship Experience

- **Faraz Hashempoor: CTF High School Summer Intern: Arizona State University** *Summer 2025*
  Conducted a semi-automated analysis of safety and privacy conversations on ChatGPT-based mental health support. [Poster]
- **Jacob Li: CTF High School Summer Intern: Arizona State University** *Summer 2025*
  Conducted a reddit-based analysis of PowerSchool data breach understanding stakeholder-specific insights and reactions. [Poster]
- **Shreyan Nath: CTF High School Summer Intern: Arizona State University** *Summer 2025*
  Conducted a mixed methods reddit-based analysis on users' safety and privacy concerns around virtual reality technologies. [Poster]
- **Riya Dhuri: CTF High School Summer Intern: Arizona State University** *Summer 2024*
  Conducted a retrospective study exploring young adults' maturation toward social media safety and privacy. [Poster]
- **Deepika Moola: CTF High School Summer Intern: Arizona State University** *Summer 2024*
  Conducted a mixed methods reddit-based analysis on how people on the internet talk about online safety and privacy. [Poster]

Skills

- **Programming Languages:** Python, C/C++, Bash, R
- **Qualitative Analysis:** MaxQDA, Google Sheets, ReCal2
- **AI & LLM Frameworks:** LangChain, LangGraph, Ollama, Hugging Face (Transformers), OpenClaw
- **AI Protocols:** MCP, A2A
- **Other Tools & Technologies:** VirtualBox, Ubuntu Server, Git, Docker, OpenLDAP, NFS, Slurm